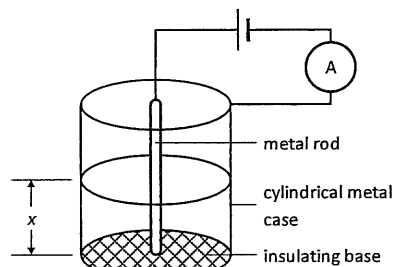


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/21]

An electrical circuit is connected to a cylindrical metal case with an insulating base as shown. Conducting liquid fills the case to a depth of x .



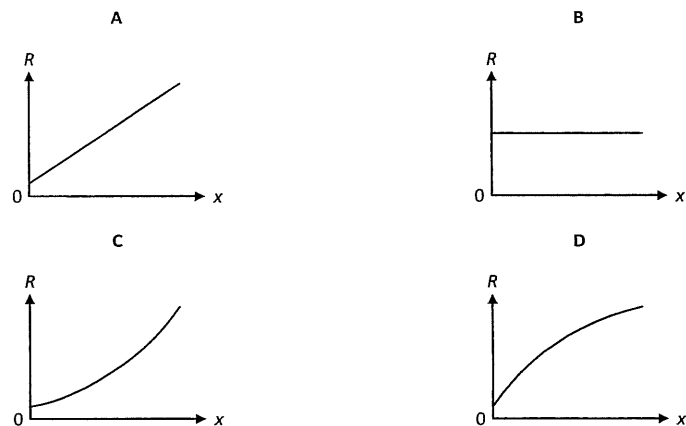
How does the current I in the ammeter vary as x changes?

- A $I \propto x$ B $I \propto x^2$ C $I \propto \frac{1}{x}$ D $I \propto \frac{1}{x^2}$

2. [AJC/H2/Prelims 2018/P1/17]

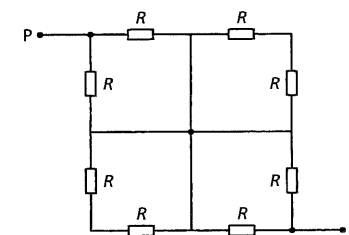
When a metal wire is stretched, it becomes longer.

Which graph best represents the variation with extension x of the resistance R of the wire?



3. [AJC/H2/Prelims 2018/P1/18]

Eight identical resistors, each of resistance R , are connected in a network as shown below.

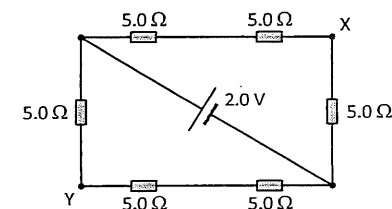


What is the effective resistance between the terminals P and Q?

- A $\frac{R}{8}$ B $\frac{R}{2}$ C R D $2R$

4. [CJC/H2/Prelims 2018/P1/22]

Six 5.0Ω resistors are connected to a 2.0 V cell of negligible internal resistance, as shown in the figure below.

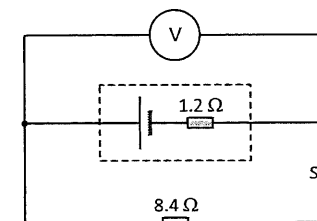


The potential difference between X and Y is

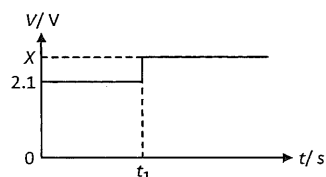
- A 0 V B $\frac{2}{3} \text{ V}$ C $\frac{8}{9} \text{ V}$ D $\frac{4}{3} \text{ V}$

5. [DHS/H2/Prelims 2018/P1/21]

A cell with internal resistance 1.2Ω is connected in the circuit as shown.



The graph shows the variation with time t of the voltmeter reading V . At time $t = 0 \text{ s}$, switch S is closed. At time $t = t_1$, switch S is opened and a rise in the voltmeter reading V was observed.

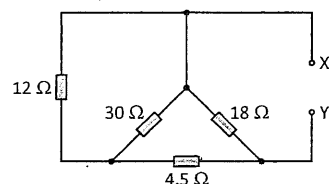


What is the value of X ?

- A 2.2 V B 2.4 V C 3.6 V D 4.2 V

6. [DHS/H2/Prelims 2018/P1/22]

The circuit diagram shows a network of resistors.

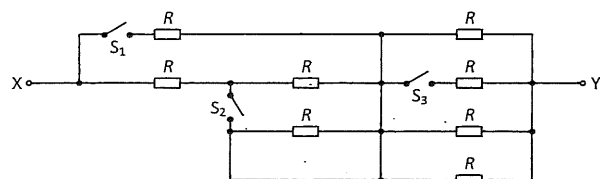


What is the effective resistance between the points X and Y?

- A 3.5 Ω B 7.6 Ω C 10.5 Ω D 15.0 Ω

7. [HCI/H2/Prelims 2018/P1/20]

A network is constructed using eight resistors, each of resistance R , and three switches S_1 , S_2 and S_3 .



Which of the following combination will give rise to the minimum total resistance between points X and Y?

	S_1	S_2	S_3
A	closed	closed	closed
B	closed	open	closed
C	open	closed	closed
D	open	open	open

8. [HCI/H2/Prelims 2018/P1/21]

A row of 30 decorative lights, connected in series, is connected to a mains transformer. When the supply is switched on, the lights do not work. The owner uses a voltmeter to test the circuit. When the voltmeter is connected across the fifth bulb in the row, a reading of zero is obtained.

Which of the following scenarios described is **not** possible?

- A Only the filament of the fifth bulb has broken.
 B The fuse in the mains transformer has blown.
 C The filament of at least one of the other bulbs has broken.
 D There is a break in the wire from the supply to the transformer.

9. [IJC/H2/Prelims 2018/P1/20]

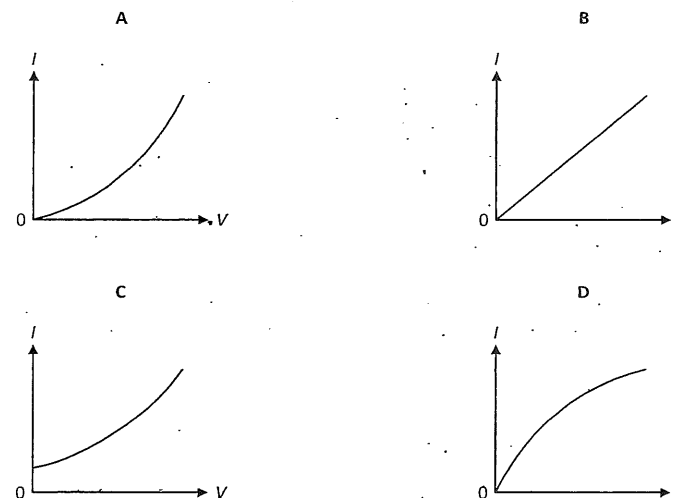
A potential difference of 12 V is applied across a resistor for a time interval of 5.0 s. The current flowing through the resistor is 3.0 A.

Which of the following statements is incorrect?

- A The resistance of the resistor is 4.0 Ω.
 B The energy dissipated in the resistor is 36 J.
 C The charge passing through the resistor is 15 C.
 D The potential difference across the resistor is 12 J C⁻¹.

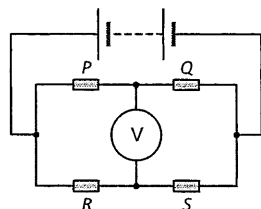
10. [IJC/H2/Prelims 2018/P1/21]

Which of the following graphs best represents the relationship between the current I and applied p.d. V for a thermistor?



11. [JJC/H2/Prelims 2018/P1/22]

The circuit diagram shows four resistors of different resistances P , Q , R and S connected to a battery.



The voltmeter reading is zero.

Which equation is correct?

- A $P - Q = R - S$ B $P - S = Q - R$ C $PQ = RS$ D $PS = QR$

12. [JJC/H2/Prelims 2018/P1/16]

A p-n junction diode has the forward current-p.d. characteristic as shown in Fig. A. It is connected in series with a variable, low voltage d.c. power supply, a meter of negligible internal resistance and a $50\ \Omega$ resistor as shown in Fig. B.

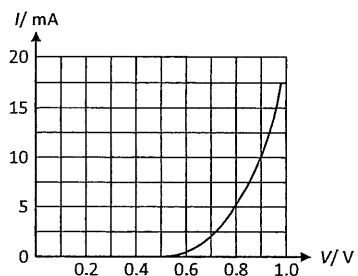


Fig. A

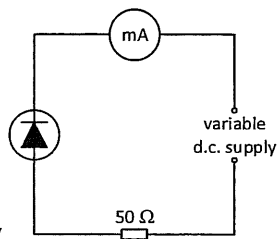


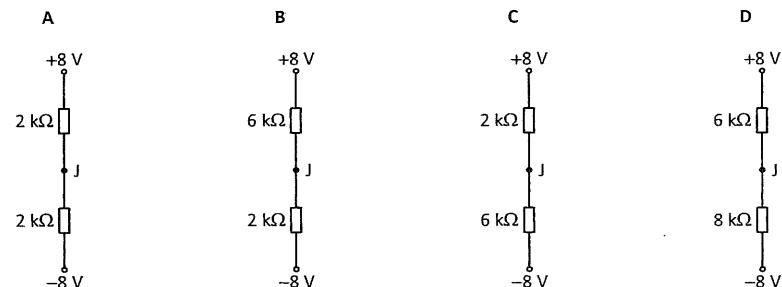
Fig. B

When the meter reads 5 mA, the potential difference across the supply is about

- A 0.25 V
B 0.80 V
C 1.05 V
D 1.25 V

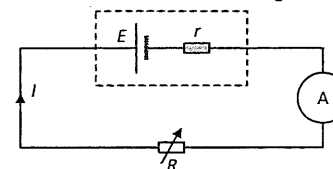
13. [JJC/H2/Prelims 2018/P1/17]

In which of the circuits below is the potential at J equal to -4 V ?



14. [MI/H2/Prelims 2018/P1/23]

A battery of e.m.f. E and internal resistance r delivers a current I through a variable resistor R as shown below.



The table below shows two different values of R and corresponding currents I measured using an ammeter of negligible resistance.

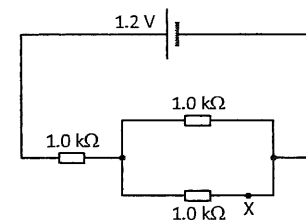
R/Ω	I/A
1.0	3.0
2.0	2.0

What is the value of e.m.f. E of the battery?

- A 3.0 V B 3.5 V C 4.0 V D 6.0 V

15. [MJC/H2/Prelims 2018/P1/20]

In the circuit below, 3 identical resistors of resistance $1.0\text{ k}\Omega$ are connected to a cell of 1.2 V with negligible internal resistance as shown.

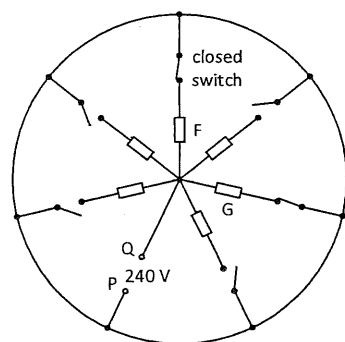


How many electrons pass through point X in a minute?

- A 2.5×10^{15} B 1.5×10^{17} C 2.5×10^{18} D 1.5×10^{20}

16. [MJC/H2/Prelims 2018/P1/21]

Electrical sockets in a house are connected to a circuit called a ring main. The circuit is connected between P and Q to the 240 V power supply as shown.



Two devices, F and G, are currently switched on. They have resistances of $1200\ \Omega$ and $1700\ \Omega$ respectively.

What is the current supplied by the power supply and total power dissipated by both devices?

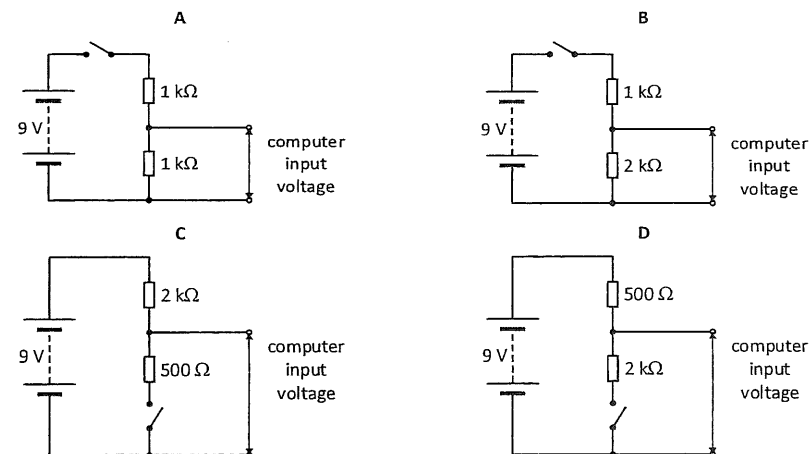
	current / A	total power dissipated / W
A	0.083	20
B	0.083	82
C	0.34	20
D	0.34	82

17. [NJC/H2/Prelims 2018/P1/19]

A computer is used to detect the change of position of a switch. To detect the change of position, the computer requires a potential difference (p.d.) of 0 V to its input at one switch position and a p.d. between 5 V and 7 V for the other switch position.

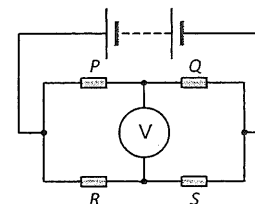
Which of the following circuits provides an input voltage to the computer that enables it to detect the change of position of the switch?

[The battery and computer in each circuit has negligible internal resistance and infinite resistance respectively.]



18. [NJC/H2/Prelims 2018/P1/20]

The diagram shows a circuit with four resistors with resistance P, Q, R and S connected to a battery.

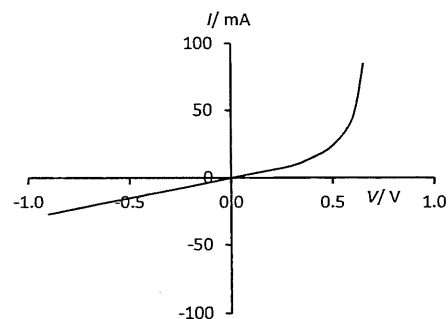


The voltmeter reading is zero. Which of the following equations is correct?

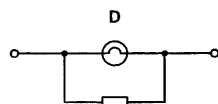
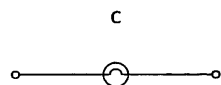
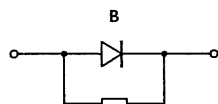
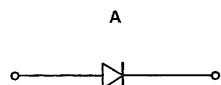
- A $P - Q = R - S$
 B $P - S = Q - R$
 C $PQ = RS$
 D $PS = QR$

19. [NYJC/H2/Prelims 2018/P1/20]

A student is given a sealed box containing a concealed electrical circuit. The figure shows the current-voltage characteristics of the box.

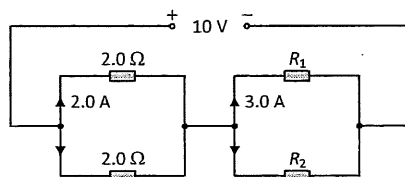


Which circuit is most likely to be enclosed within the box?



20. [NYJC/H2/Prelims 2018/P1/21]

Four resistors are connected across a 10 V supply as shown in the circuit below.

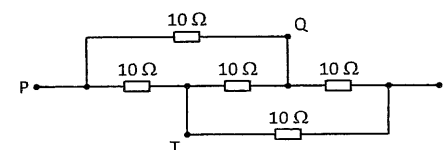


What are the resistances R_1 and R_2 ?

	R_1	R_2
A	1.0 Ω	3.0 Ω
B	2.0 Ω	2.0 Ω
C	2.0 Ω	6.0 Ω
D	3.0 Ω	3.0 Ω

21. [PJC/H2/Prelims 2018/P1/19]

A network of resistors each of resistance 10Ω are shown.

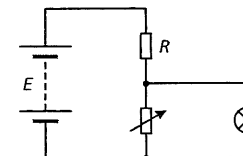


What is the effective resistance between Q and T?

- A 2.0 Ω B 5.0 Ω C 10 Ω D 12 Ω

22. [PJC/H2/Prelims 2018/P1/20]

A battery of e.m.f. E and negligible internal resistance is connected to a resistor of resistance R , a variable resistor and a lamp as shown.

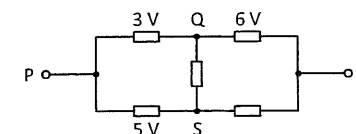


What happens to the brightness of the lamp as the resistance of the variable resistor is increased?

- A The bulb becomes dimmer.
B The bulb becomes brighter.
C The brightness remains the same.
D The bulb becomes brighter initially, and then becomes dimmer.

23. [RVHS/H2/Prelims 2018/P1/17 Modified]

There is a current flowing from P to R in the resistor network shown. The potential difference (p.d.) between P and Q is 3 V. The p.d. between Q and R is 6 V. The p.d. between P and S is 5 V.

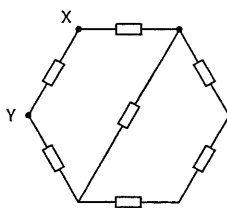


What is the p.d. between Q and S, and S and R?

	p.d. between Q and S	p.d. between S and R
A	2 V	4 V
B	2 V	10 V
C	3 V	4 V
D	3 V	10 V

24. [RVHS/H2/Prelims 2018/P1/21]

The diagram shows a network of 7 resistors, each with resistance R .

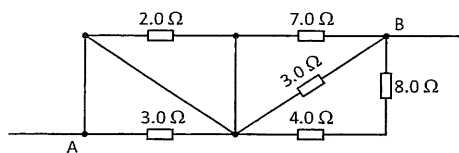


What is the resistance between points X and Y?

- A $\frac{2}{3}R$ B $\frac{3}{5}R$ C $\frac{7}{11}R$ D $\frac{11}{15}R$

25. [SRJC/H2/Prelims 2018/P1/23]

Six resistors are connected in a circuit as shown below.

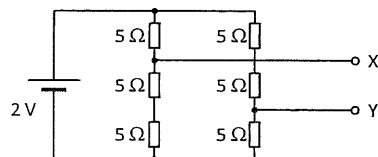


What is the effective resistance of the circuit between terminals A and B?

- A 1.2Ω B 1.8Ω C 3.0Ω D 3.4Ω

26. [TPJC/H2/Prelims 2018/P1/22]

Six resistors, each of resistance 5Ω , are connected to a 2 V cell with negligible internal resistance.

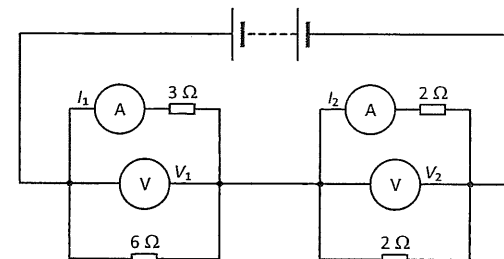


What is the potential difference between terminals X and Y?

- A $\frac{2}{3} \text{ V}$ B $\frac{8}{9} \text{ V}$ C $\frac{4}{3} \text{ V}$ D 2 V

27. [TPJC/H2/Prelims 2018/P1/23]

In the circuit shown, the ammeters have negligible resistance and the voltmeters have infinite resistance.

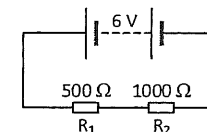


The readings on the meters are I_1 , I_2 , V_1 and V_2 , as labelled. Which is correct?

- A $I_1 > I_2$ and $V_1 > V_2$ B $I_1 > I_2$ and $V_1 < V_2$
C $I_1 < I_2$ and $V_1 > V_2$ D $I_1 < I_2$ and $V_1 < V_2$

28. [VJC/H2/Prelims 2018/P1/21]

The diagram shows a 6 V battery, with negligible internal resistance, connected in series to two resistors R_1 and R_2 of resistance 500Ω and 1000Ω , respectively.



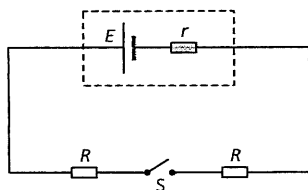
A third resistor with a resistance of 500Ω is placed in parallel across R_2 .

Which statement about the new circuit is correct?

- A The current in R_2 is larger than before.
B The current through the battery is smaller than before.
C The potential difference across R_1 is larger than before.
D The potential difference across R_2 is now greater than the potential difference across R_1 .

29. [YJC/H2/Prelims 2018/P1/20]

A battery, with an e.m.f. E and internal resistance r , is connected to a switch S and two identical resistors in series. Each resistor has resistance R .



Which of the following statements is correct when the switch S is closed?

- A The rate at which energy is dissipated across R is $\frac{E^2}{4R}$.
- B The rate at which energy is dissipated across R is $\frac{E^2 r}{2R + r}$.
- C The rate at which energy is dissipated across r is $\frac{E^2 r}{(2R + r)^2}$.
- D It is impossible to determine the rate at which energy is dissipated across R .

30. [YJC/H2/Prelims 2018/P1/21]

Wire X has resistivity ρ . Another wire Y, of the same resistance as X, has triple the length and double the diameter of wire X.

The resistivity of wire Y is

- A $\frac{2\rho}{3}$ B $\frac{3\rho}{4}$ C $\frac{4\rho}{3}$ D $\frac{3\rho}{2}$

31. [YJC/H2/Prelims 2018/P1/22]

With 4 resistors, each having a resistance of $12\ \Omega$, it is impossible to arrange all 4 resistors to have an effective resistance of

- A $9\ \Omega$ B $12\ \Omega$ C $20\ \Omega$ D $24\ \Omega$

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P2/5 Modified]

- b. Fig. 1.2 shows a fixed resistor R and a thermistor connected in parallel to a battery with an electromotive force E and internal resistance r .

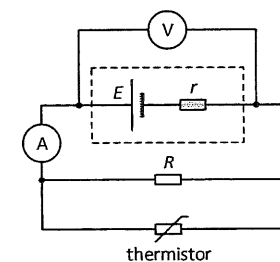


Fig 1.2

Fig. 1.3 shows how the readings in the voltmeter V changes as current drawn from the battery, I , varies.

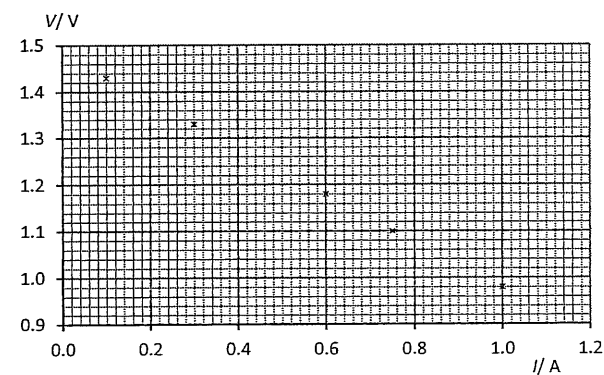


Fig. 1.3

- i. Use Fig. 1.3 to determine E and r . [3]
- ii. State and explain how the power output of R varies as temperature of the thermistor increases. [3]

[Total: 6]

2. [AJC/H2/Prelims 2018/P2/5]

- a. Fig. 2.1 shows the variation with voltage V of the current I across a filament lamp rated 6.0 V, 1.5 W.

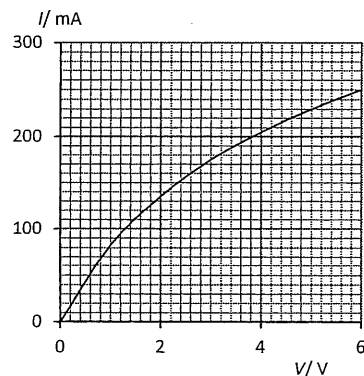


Fig. 2.1

- In microscopic terms, explain why the resistance of the filament lamp increases as V increases. [3]
 - On Fig. 2.1, draw a line to show the variation with V of I if the filament lamp is assumed to be ohmic. [1]
 - Use Fig. 2.1 to determine the change in resistance of the filament lamp when V increases from 4.0 V to 6.0 V. [2]
- b. A student designs a circuit for a night light using the filament lamp in a. and a light dependent resistor (LDR), as shown in Fig. 2.2.

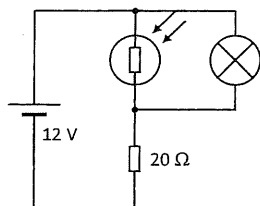


Fig. 2.2

The LDR has a resistance of $10\ \Omega$ in daylight and increases to $1000\ \Omega$ in the dark.

- Explain why the lamp will not operate at its rated power in daylight. [2]
- Calculate the resistance of the LDR in order for the lamp to operate at its rated power of 1.5 W. [3]

[Total: 11]

3. [CJC/H2/Prelims 2018/P3/8 Part]

- a. A cell of e.m.f. E and internal resistance r is connected in series to a resistor of resistance R .
- Define *electromotive force*. [1]
 - Show, by considering energy conversion, that V , the terminal potential difference of the cell is $V = E - Ir$ where I is the current flowing in the cell. [2]
- b. Fig. 3.1 shows a battery with e.m.f. E and an internal resistance r connected to a uniform nichrome resistance wire MN. J is a movable jockey which can slide along wire MN. The voltmeter and the ammeter are taken to be ideal.

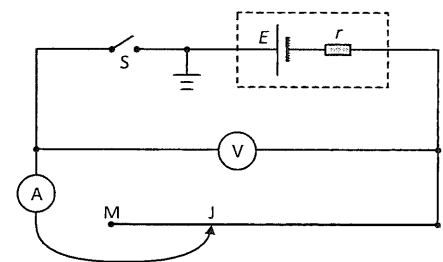


Fig. 3.1

The voltmeter readings V and ammeter readings I obtained for different lengths of JN are used to plot the graph in Fig. 3.2.

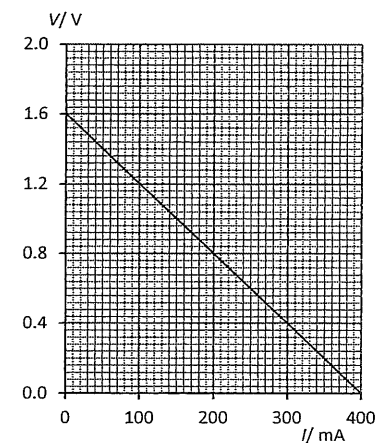


Fig. 3.2

- Deduce from Fig. 3.2, the e.m.f. E and the internal resistance r of the cell. [3]
- J is placed at the position such that maximum power is delivered from the cell to the wire JN. Determine the potential at J and N. [3]

- iii. On Fig. 3.3 sketch the graph to show how electric potential varies with position along the wire JN. Label the vertical axis clearly. Explain your answer. [3]

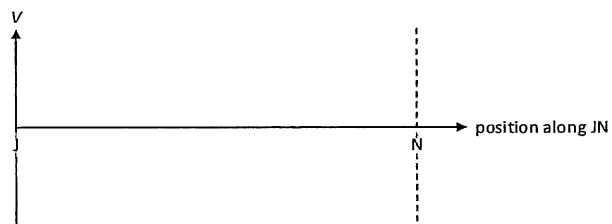


Fig. 3.3

- iv. Suggest, with a reason, why the position of the jockey J has to be shifted, when the battery has been used for a prolonged period of time for maximum power transfer. [1]

[Total: 10]

4. [DHS/H2/Prelims 2018/P3/7 (part)]

- a. Distinguish between the electromotive force and the potential difference in terms of energy considerations. [2]
b. Felix connects a voltmeter, of resistance R_V , and an ammeter, of resistance R_A , as shown in Fig. 4.1 to measure the resistance R of a resistor. V is the voltmeter reading, I is the ammeter reading and E is the e.m.f. of the cell.

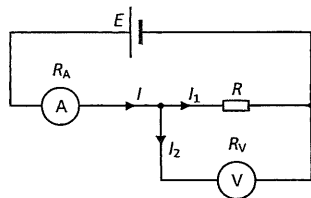


Fig. 4.1

- i. Derive an expression for the value of R in terms of I , V and R_V . [2]
ii. Felix rearranges the circuit as shown in Fig. 4.2 to measure the same resistance R .

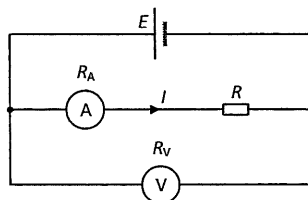


Fig. 4.2

Derive an expression for the value of R in terms of I , V and R_A . [2]

- iii. Hence, suggest what the values of R_V and R_A should be such that the value of R is equal to V/I . [2]

- d. During his course of study in Physics, Felix comes across an electrical component. The variation with potential difference V of current I for the component is shown in Fig. 4.4.

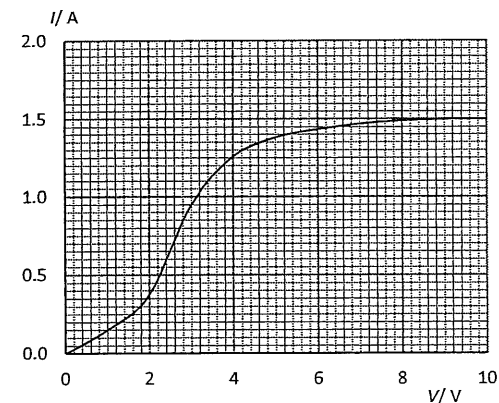


Fig. 4.4

- i. Use Fig. 4.4 to determine the resistance of the component at 3.4 V. [1]
ii. Describe how the resistance of this component changes with the potential difference applied across it from 0 V to 6.0 V. [2]

[Total: 12]

5. [HCI/H2/Prelims 2018/P2/5 (part)]

- a. The variation with temperature T of the resistance R_T of a thermistor is shown in Fig. 6.1.

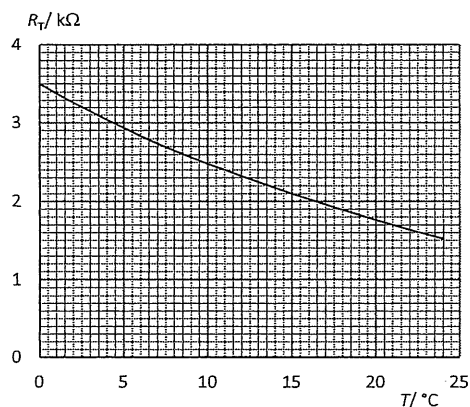


Fig. 6.1

The thermistor is connected in series with a resistor R as shown in the circuit in Fig. 6.2.

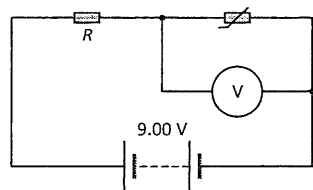


Fig. 6.2

The battery has e.m.f. 9.00 V and negligible internal resistance. The voltmeter has infinite resistance.

- For the thermistor at 22.5 °C, determine the resistance of the thermistor.
- Given that the voltmeter reading is 2.70 V, determine the resistance of resistor R .

[1]

[2]

6. [IJC/H2/Prelims 2018/P2/4]

In order to investigate the variation of the current I in a variable resistor with the potential difference V across it, a student set up the following circuit.

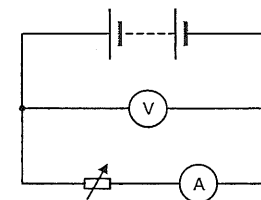


Fig. 7.1

The variation with I of V is shown below.

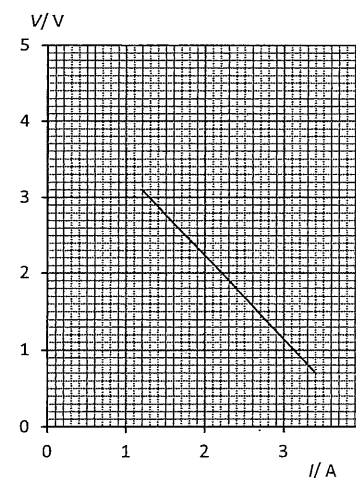


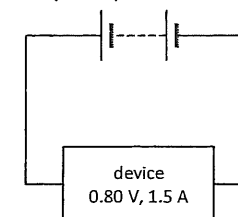
Fig. 7.2

- With reference to the graph, explain why the internal resistance of the battery is not negligible.
 - Show that the e.m.f of the battery and its internal resistance is 4.4 V and 1.1 Ω respectively.
- The battery in a. is to be used as the power source for an electrical device. The device is rated as 0.80 V, 1.5 A. Complete the circuit below to show how the battery may be connected so that the device operates normally. Calculate the value of any other component you may use.

[1]

[2]

[3]



[Total: 6]

7. [JJC/H2/Prelims 2018/P3/6 Modified]

- a. The resistivity of copper is $1.7 \times 10^{-8} \Omega \text{ m}$. Show that the resistance of a copper wire of length 1.5 m and cross-sectional area of $3.2 \times 10^{-9} \text{ m}^2$ is about 8.0Ω . [1]
- b. The copper wire in (a) is used to connect a circuit as shown in Fig. 8.1. Cell A has an e.m.f of 12.0 V and internal resistance 1.0Ω . A 2.0Ω resistance is used and connected in series with cell A.

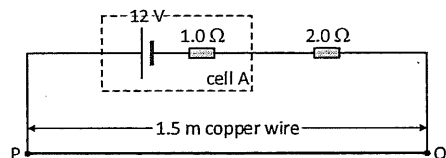


Fig. 8.1

Determine the drift velocity of the electrons flowing in the copper if the number of electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$. [2]

8. [JJC/H2/Prelims 2018/P3/6 Modified]

A small electric torch is powered by a single cell which supplies 1.60 J of energy per coulomb of charge passing through the cell. When the torch is switched on, the cell supplies a constant current of 0.50 A to bulb X. The potential difference across the bulb is 1.20 V.

- a. Show that the internal resistance, r , of the cell is 0.80Ω . [2]
- b. The bulb X is replaced by another bulb Y, which draws a current of 0.30 A. Calculate the potential difference across bulb Y. [2]
- c. Calculate the power lost in the cell when connected to bulb Y. [2]
- d. When bulb Y is replaced by bulb Z, the power drawn from the cell is maximum. Determine the maximum current that can be drawn from the cell. [2]

[Total: 8]

9. [JJC/H2/Prelims 2018/P3/7]

- a. A copper pipe has an internal diameter of 4.00 cm and an external diameter of 7.00 cm. The resistivity of copper is $1.68 \times 10^{-8} \Omega \text{ m}$. Determine the resistance of 15.0 m of this pipe. [3]
- b. A number of resistors are connected to a 6.0 V cell in an electrical circuit as shown in Fig. 10.1.

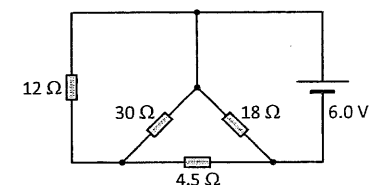


Fig. 10.1

With reference to Fig. 10.1,

- i. show that the equivalent resistance of the circuit is 7.6Ω . [1]
- ii. determine the current in the 18Ω resistor. [1]
- iii. determine the power dissipated in the 4.5Ω resistor. [3]

[Total: 8]

10. [MI/H2/Prelims 2018/P3/4 Modified]

Fig. 11.1 shows how the resistance R of a light-dependent resistor (LDR) varies with the intensity I of the light incident on it.

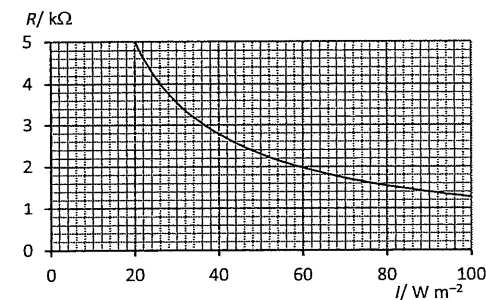


Fig. 11.1

Fig. 11.2 shows a light-sensing potential divider circuit used in a lamp where the potential difference across the LDR can be used to control the brightness of the lamp in a room.

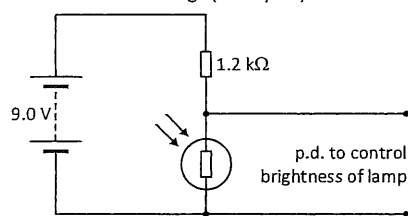


Fig. 11.2

The battery has an e.m.f. of 9.0 V and negligible internal resistance. A fixed resistor of $1.2 \text{ k}\Omega$ and the LDR is connected in series with the battery. When the room is in a low-light condition, the potential difference across the LDR reaches 7.0 V.

- State and explain quantitatively, if the resistance of the LDR is inversely proportional to the intensity of the light incident on it. [1]
- Calculate the resistance R of the LDR. [3]
- Use Fig. 11.1 to determine the intensity of light incident on the LDR. [1]
- Fig. 11.3 shows a close-up of the LDR device used in the circuit in Fig. 11.2. The LDR consists of a uniform strip of semiconductor whose resistance is dependent on the intensity of the light incident on it. The cross-sectional area of the strip is 0.50 mm^2 .

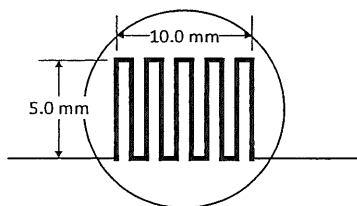


Fig. 11.3

Use your answer in b. to estimate the resistivity of the LDR.

[2]

11. [MJC/H2/Prelims 2018/P3/7 (part)]

- A power bank (which is basically a battery) can be used to power many devices at the same time. A power bank of e.m.f. 12.0 V and internal resistance 3.0Ω is connected to multiple devices in the circuit shown in Fig. 12.1.

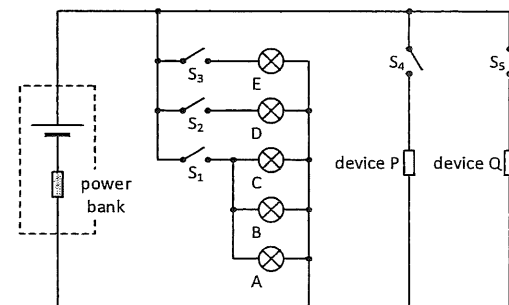


Fig. 12.1

The power bank is connected to 5 identical lamps (A, B, C, D and E) and 2 devices (P and Q). The lamps and devices can be turned on and off using the various switches (S_1 , S_2 , S_3 , S_4 and S_5).

- Explain what is meant by “e.m.f. of 12.0 V” with reference to the power bank. [2]
- State the effect of closing switch S_1 . [1]
- All the switches are now closed. Given the data below, calculate the current supplied by the power bank.
Resistance of each lamp = 25.0Ω
Resistance of device P = 38.0Ω
Resistance of device Q = 42.0Ω [2]
- Calculate the terminal potential difference of the power bank when all the switches are closed. [2]
- State and explain the effect, if any, on the brightness of the lamps if switches S_4 and S_5 are now opened while the rest remain closed. [2]

12. [NIC/H2/Prelims 2018/P2/4]

- a. The I - V characteristics of resistor R and resistor X are investigated. The results obtained are shown in Fig. 13.1.

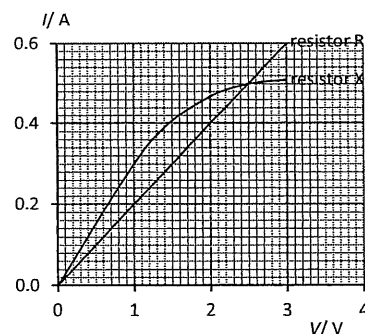


Fig. 13.1

- i. Student 1 sets up the circuit with resistors R and X in series across an unknown e.m.f. source with negligible internal resistance as shown in Fig. 13.2.



Fig. 13.2

Determine the value of the unknown e.m.f. source if the current through resistor X is 0.22 A. [2]

- ii. Student 2 sets up the circuit using a variable resistor, resistors R and X as shown in Fig. 13.3 using the same e.m.f. source.

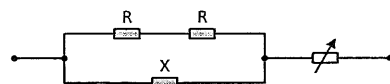


Fig. 13.3

Determine the resistance of the variable resistor given that the ratio of $\frac{\text{resistance of X}}{\text{resistance of R}} = 0.80$. [3]

- iii. Student 2 modifies his circuit by adding an ideal ammeter and a voltmeter as shown in Fig. 13.4.

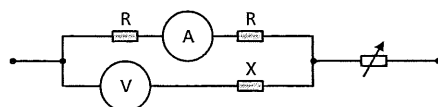


Fig. 13.4

Explain if this circuit is appropriate for determining the graphs over the range of V and I as shown in Fig. 13.1. [3]

- b. A student uses the equation $I = nAve$ to determine the drift velocity of the electrons, v , in the wire. State, with a reason, whether this is the maximum velocity of the free electrons in the wire. [2]

[Total: 10]

13. [NIC/H2/Prelims 2018/P3/3 Part]

- a. A battery of e.m.f. 4.50 V and negligible internal resistance is connected to a fixed resistor of resistance 1200Ω and a thermistor, as shown in Fig. 14.1.

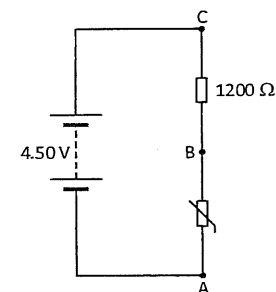


Fig. 14.1

At room temperature, the thermistor has a resistance of 1800Ω .

A uniform resistance wire PQ of length 1.000 m is now connected to the resistor and the thermistor, as shown in Fig. 14.2.

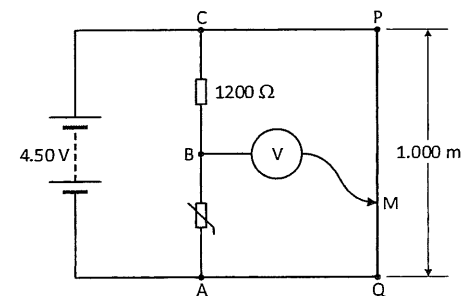


Fig. 14.2

A voltmeter is connected between point B and a movable contact M on the wire.

- i. Explain why, for constant current in the wire, the potential difference between any two points on the wire is proportional to the distance between the points. [2]
- iii. The temperature of the thermistor increases slightly during the measurement. Explain the effect on the length of wire between M and Q for the voltmeter to show zero reading. [2]

[Total: 6]

14. [NYJC/H2/Prelims 2018/P2/6]

Fig. 15.1 shows a circuit consisting of two resistors connected in series to a d.c. supply.

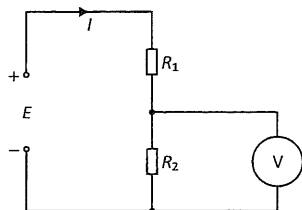


Fig. 15.1

The resistors have resistances R_1 and R_2 . The supply has e.m.f. E and negligible internal resistance. The current from the supply is I . The voltmeter has an infinite resistance.

- a. By reference to Ohm's Law, show that the voltmeter reading V is given by the relation

$$V = \left(\frac{R_2}{R_1 + R_2} \right) E \quad [2]$$

- b. Fig. 15.2 shows a circuit that includes a thermistor.

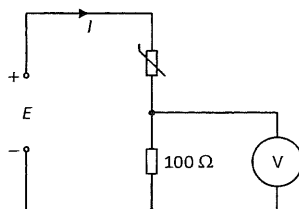


Fig. 15.2

Describe and explain how the voltmeter reading changes as the temperature of the thermistor is increased. [2]

- c. Fig. 15.3 shows a circuit that includes a potentiometer wire. It is used to test E_T , the e.m.f. of the test cell and the balance length is found to be b .

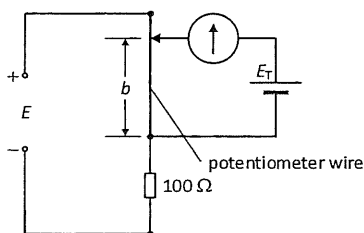


Fig. 15.3

- i. The potentiometer wire, of 1.000 m long, is made of nichrome wire of resistivity $1.57 \times 10^{-6} \Omega \text{ m}$ and has a uniform radius of 0.10 mm. Determine the resistance of the potentiometer wire. [2]

[Total: 2]

15. [PJC/H2/Prelims 2018/P3/6]

A filament lamp is connected in series to a cell of e.m.f. 4.0 V and negligible internal resistance and a variable resistor as shown in Fig. 17.1. The variation with the potential difference V across the filament lamp of the current I through the lamp is shown in Fig. 17.2.

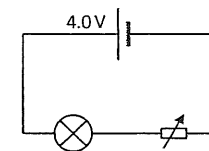


Fig. 17.1

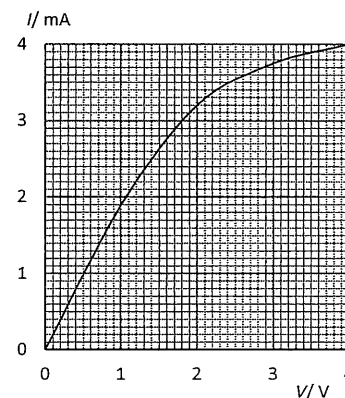


Fig. 17.2

- a. Determine the resistance of the variable resistor when V is 1.4 V. [2]
- b. On Fig. 17.3, sketch a labelled graph of the resistance R of the filament lamp against V between $V = 1.0 \text{ V}$ and 4.0 V . [2]

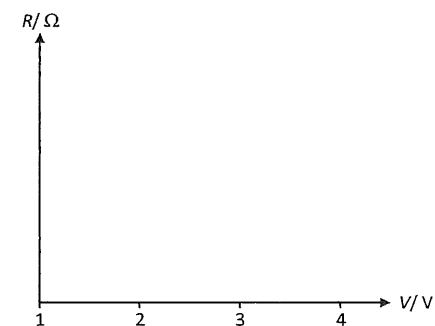


Fig. 17.3

- c. Discuss how the resistance of a filament lamp changes with temperature. [2]

[Total: 6]

16. [RVHS/H2/Prelims 2018/P2/5]

- a. Distinguish between electromotive force (e.m.f.) and potential difference (p.d.) using energy considerations. [2]
- b. In order to determine the internal resistance r of a 12.00 V battery, a student sets up the circuit shown in Fig. 18.1. The variable resistor CD has a maximum resistance of $1000\ \Omega$.

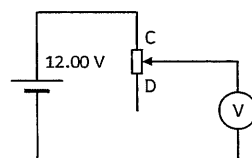


Fig. 18.1

It is found that when the sliding contact of the variable resistor is placed at C, the digital voltmeter reading is 11.99 V. When the sliding contact is moved down from C to D, the digital voltmeter reading drops from 11.99 V to 11.00 V.

- i. State the potential difference across the internal resistance of the battery when the sliding contact of the variable resistor is placed at C. [1]
- ii. Explain why the presence of internal resistance in the battery reduces its output power. [1]
- c. Using the results of b.i., show that $R = 1199r$ where R is the resistance of the voltmeter and r is the internal resistance. [1]

17. [RVHS/H2/Prelims 2018/P2/7 Modified]

- a. In order to light up a set of lamps, each having a resistance of $20\ \Omega$, a student sets up a circuit as shown in Fig. 19.1.

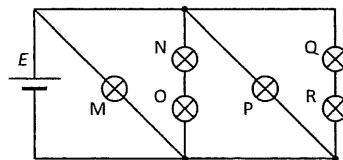


Fig. 19.1

- i. Determine the effective resistance of the circuit. [1]
- ii. State and explain the relative brightness of the lamps. [2]
- b. In another circuit shown in Fig. 19.2, a uniform wire AB of length 0.80 m and resistance $1.5\ \Omega$ is connected in series with a resistor of $5.0\ \Omega$ and a cell of e.m.f. 9.0 V with internal resistance $1.0\ \Omega$.

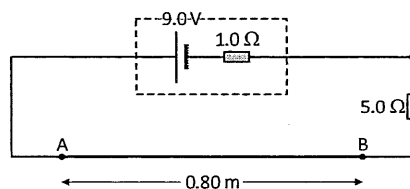


Fig. 19.2

- i. Calculate the potential difference across wire AB. [2]

18. [SRJC/H2/Prelims 2018/P3/9]

- b. A circuit is set up as shown in Fig. 20.1.

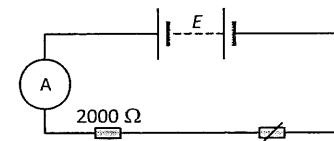


Fig. 20.1

The battery, of emf E and negligible internal resistance provides $2.4 \times 10^5\text{ J}$ of electrical energy to the $2000\ \Omega$ resistor and thermistor when a charge of $2.2 \times 10^4\text{ C}$ passes through the ammeter. At room temperature, the thermistor has a resistance of $1800\ \Omega$.

- i. Sketch on Fig. 20.2 the variation with temperature θ of resistance R in a thermistor. [1]



Fig. 20.2

- ii. For the thermistor at room temperature,
- show that E is 11 V, [1]
 - determine the time taken for a charge of $2.2 \times 10^4\text{ C}$ to pass through the ammeter, [2]
 - show that the fraction of power dissipated in the thermistor is 0.47. [1]

- c. A uniform resistance wire PQ of length 1.2 m is subsequently connected across the resistor and thermistor, as shown in Fig. 20.3. A sensitive voltmeter is connected between point Y and a moveable contact M on the wire.

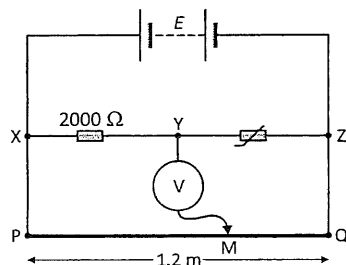


Fig. 20.3

- At room temperature, the contact M is moved along PQ until the voltmeter shows zero reading. Calculate the length of wire between M and Q. [2]
- State and explain the effect, if any, on the length of the wire between M and Q for the voltmeter to show zero reading if each of the following changes takes place independently. [2]
 - The thermistor is warmed slightly. [2]
 - A uniform wire with a bigger cross-sectional area is used to replace PQ. [2]

19. [TPJC/H2/Prelims 2018/P2/2]

- a. A uniform wire AB of length 100 cm is connected between the terminals of a cell of e.m.f. 1.5 V and negligible internal resistance, as shown in Fig. 21.1.

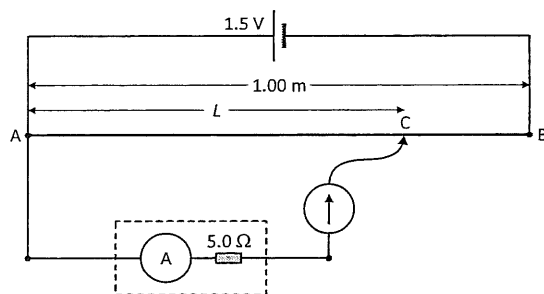


Fig. 21.1

An ammeter of internal resistance 5.0Ω is connected to end A of the wire and to a contact C that can be moved along the wire.

Determine the reading on the ammeter for the contact C placed

- at A, [1]
- at B. [1]

- b. Using the circuit in Fig. 21.1, the ammeter reading I is recorded for different distances L of the contact C from end A of the wire. Some data points are shown on Fig. 21.2.

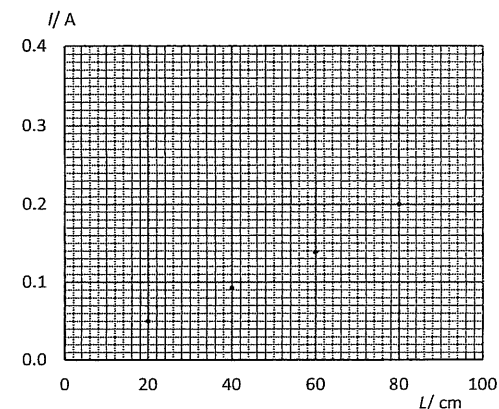


Fig. 21.2

- Use your answers in a. to plot data points on Fig. 21.2 corresponding to the contact C placed at end A and at end B of the wire. [2]
 - Draw a line of best fit for all of the data points and hence determine the ammeter reading for contact C placed at the midpoint of the wire. [2]
 - Use your answer in b.ii. to calculate the potential difference between A and the contact C for the contact placed at the midpoint of AB. [2]
- c. Explain why, although the contact C is at the midpoint of wire AB, the answer in b.iii. is **not** numerically equal to half of the e.m.f. of the cell. [2]

[Total: 10]

20. [VJC/H2/Prelims 2018/P3/9]

- a. i. Define the *ohm*. [1]
- ii. On Fig. 22.1 below, sketch a graph to show how you would expect the current in a tungsten filament lamp to change as the potential difference across the lamp increases slowly from $-V$ to $+V$. [1]

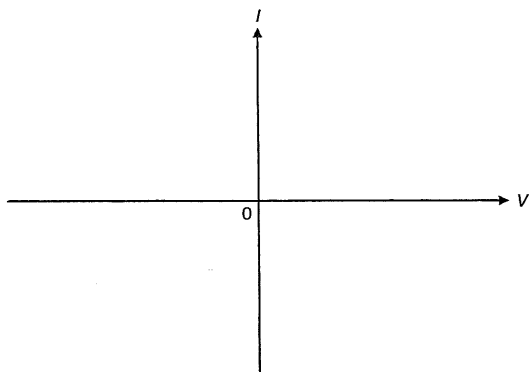


Fig. 22.1

- iii. State how you would obtain the resistance of the lamp from your graph at a particular potential difference across the lamp. [1]
- iv. Draw a circuit that could be used to investigate the I - V characteristics of a lamp as shown in Fig. 22.1. [2]
- b. A circuit is setup in Fig. 22.2. A cell E with internal resistance r is connected to a network of resistors.

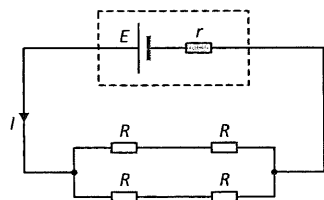


Fig. 22.2

- i. Determine the current I in the circuit, in terms of R , r and E . [2]
- ii. Show that the fraction of power dissipated in the external resistors by the battery is $\frac{R}{R+r}$. [2]
- iii. Suggest a possible reason for having 4 resistors connected in a network in Fig. 22.2 instead of just using a single resistor of resistance R . [1]
- c. A resistor of resistance of $8.2 \text{ k}\Omega$ and a LDR (light dependent resistor) are connected in series, as shown in Fig. 22.3. The resistance of the LDR when light is shone on it is 420Ω and when dark is $134 \text{ k}\Omega$.

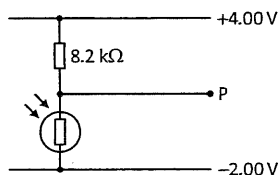


Fig. 22.3

Calculate the potential at point P when

- i. light is shone on the LDR, [2]
- ii. no light is shone on the LDR (dark). [1]
- iii. An alarm is now connected in parallel with the LDR in Fig. 22.3 and a laser beam is directed at the LDR in a simple burglar alarm system. When a burglar blocks the laser that is shining continuously on the LDR, the alarm is triggered. Assume that the resistance of alarm is very large. Explain clearly how the LDR is used to turn on/off the alarm in such a system. [2]

21. [VJC/H2/Prelims 2018/P2/5 Part Modified]

- a. Fig. 23.1 shows how the resistance R of a metal resistor and the resistance R_T of a thermistor change with temperature.

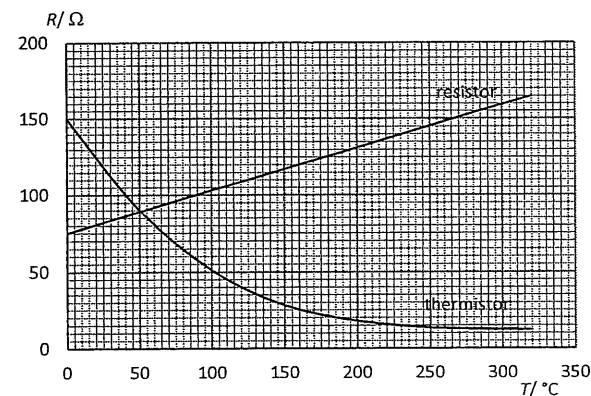


Fig. 23.1

- i. State the values of the resistance R and R_T at a temperature of 105°C . [2]
- ii. The resistor and thermistor are connected in series to a 12.0 V battery of negligible internal resistance, as shown in Fig. 23.2.

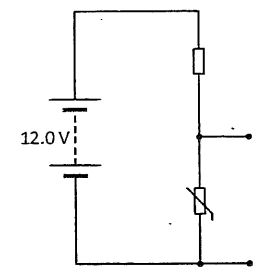


Fig. 23.2

- Calculate the potential difference across XY at 105°C . [2]
- iii. Assuming that the temperature of the resistor always equals the temperature of the thermistor, deduce the temperature, without any further calculations when the potential difference across the resistor is 6.0 V . Explain your answer. [2]

